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fastino / gliner2-privacy-filter-PII-multi

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Token Classification | GLiNER2 | Safetensors | synthetic | 7 languages | extractor | pii | ner | privacy | redaction | information-extraction | span-extraction | arxiv:2605.09973 | arxiv:2604.09791 | License: apache-2.0

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Model card | Files xet | Community 1

arXiv 2605.07982 | Deploy GLiNER2 PII

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GLiNER2-PII: Multilingual PII Detection & Masking

GLiNER2-PII is a fine-tune of the GLiNER2 model (205M parameters) for detecting and masking personally identifiable information across **42 entity types** and **7 languages**.

Trained entirely on a constraint-driven synthetic corpus of 4,910 annotated texts, it achieves the **highest span-level F1 (0.477)** on the SPY benchmark among four compared systems — including OpenAI Privacy Filter, NVIDIA GLiNER-PII, and urchade/gliner_multi_pii-v1.

📄 Technical Report
🔗 GitHub

Downloads last month
88,014

Safetensors

Model size | 0.3B params
Tensor type | F32
Files info

Inference Providers NEW

Token Classification

This model isn't deployed by any Inference Provider.

Ask for provider support

Model tree for fastino/gliner2-p...

Adapters — 1 model
Finetunes — 2 models
Quantizations — 4 models

Space using fastino/gliner2-p... 1

helmutz3m0/pii-detector

Quick Start

```
pip install gliner2
```

```
from gliner2 import GLiNER2

model = GLiNER2.from_pretrained("fastino/gli

text = "Email john.smith@acme.com or call +1
labels = ["email", "phone_number", "person"]

result = model.extract_entities(
    text,
    labels,
    threshold=0.5,
    include_confidence=True,
    include_spans=True,
)

print(result)
```

You can pass **any subset** of the 42 supported labels — the model conditions on the labels you provide at inference time.

Supported PII Labels (42 types)

Group	Labels
Person / names	`person`, `full_name`, `first_name`, `middle_name`, `last_name`, `date_of_birth`
Contact / address	`email`, `phone_number`, `address`, `street_address`, `city`,

Group	Labels
	state_or_region, postal_code, country
Government / tax IDs	government_id, national_id_number, passport_number, drivers_license_number, license_number, tax_id, tax_number
Banking / payment	bank_account, account_number, routing_number, iban, payment_card, card_number, card_expiry, card_cvv
Digital identity	username, ip_address, account_id, sensitive_account_id
Secrets / credentials	password, secret, api_key, access_token, recovery_code
Sensitive dates	sensitive_date, document_date, expiration_date, transaction_date

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Benchmark Results (SPY)

Evaluated on the SPY benchmark (Savkin et al., 2025) with exact-match span-level metrics:

Model	Legal P	Legal R	Legal F1	Medic P
fastino/gliner2-pii-v1	.346	**.750**	**.473**	.369

Model	Legal P	Legal R	Legal F1	Medic P
nvidia/gliner-PII	.343	.452	.390	.368
urchade/gliner_multi_pii-v1	**.467**	.317	.377	**.518**
openai/privacy-filter	.242	.656	.354	.287

Key takeaways

- **Highest F1** on both legal and medical domains.
- **Best recall** among GLiNER-based detectors (0.718 avg) — critical for redaction workflows where missed spans are data leaks.
- Consistent performance across domains (< 2-point F1 difference).

When to Use This Model

Use case	Why GLiNER2-PII
PII redaction / masking	High recall minimises missed sensitive spans
Data governance & GDPR/CCPA compliance	42 fine-grained types enable policy-specific routing
Training-data hygiene	Exact character spans for precise masking before model training
Multi-language pipelines	Trained on EN, FR, ES, DE, IT, PT, NL formats

Redaction Example

```
def redact(text, labels, threshold=0.5):
    model = GLiNER2.from_pretrained("fastino
    result = model.extract_entities(
        text, labels, threshold=threshold,
        include_spans=True,
    )
    entities = result.get("entities", {})
    spans = []
    for label, values in entities.items():
        for value in values:
            start = text.find(value)
            if start != -1:
                spans.append((start, start +

    spans.sort(key=lambda s: s[0], reverse=T
    redacted = text
    for start, end, label in spans:
        redacted = redacted[:start] + f"[{lal
    return redacted


text = "Please contact Maria Jensen at maria
labels = ["person", "email", "phone_number"]
print(redact(text, labels))
# "Please contact [PERSON] at [EMAIL] or [PH
```

Training Details

Detail	Value
Base model	GLiNER2 (205M parameters)

Detail	Value
Training data	4,910 synthetic annotated texts
PII mentions	129,951 total (mean 26.5 per example)
Generator	GPT-5.4 (temperature 0.01)
Data framework	Constraint-driven generation (same framework as Pioneer Agent)
Languages	English, French, Spanish, German, Italian, Portuguese, Dutch
Label types	42 PII entity types across 7 semantic groups

Limitations

- **Precision** (0.35–0.37 on SPY) leaves room for improvement; the model tends to over-predict `name` entities, sometimes confusing common nouns, organisation names, and product names with personal names.
- Evaluated on a **single benchmark** (SPY) covering two domains. Broader multilingual and fine-grained evaluation is ongoing.
- Training data is **fully synthetic** and has not been validated by human annotators.
- Performance on **non-European** locales and scripts has not been measured.

Improving precision

For production use, consider:

- Per-label confidence thresholds (raise threshold for `person` / `full_name`)
- Dictionary-based filtering for common false positives
- Calibration on a small domain-specific development set

Citation

```
@misc{fastino2026gliner2pii,
  title   = {GLiNER2-PII: Multilingual PII E
  author  = {{Fastino AI Team}},
  year    = {2026},
  url     = {https://huggingface.co/fastino/
}
```

Related work

```
@misc{zaratiana2026gliner2piimultilingualmod
      title={GLiNER2-PII: A Multilingual Mod
      author={Urchade Zaratiana and Ash Lewi
      year={2026},
      eprint={2605.09973},
      archivePrefix={arXiv},
      primaryClass={cs.CL},
      url={https://arxiv.org/abs/2605.09973}
}

@inproceedings{zaratiana-etal-2025-gliner2,
  title     = {GLiNER2: Schema-Driven Multi-
  author    = {Zaratiana, Urchade and Paster
  booktitle = {Proceedings of EMNLP 2025: Sy
  year      = {2025}
}
```

```
@inproceedings{zaratiana-etal-2024-gliner,
  title     = {GLiNER: Generalist Model for N
  author    = {Zaratiana, Urchade and Tomeh,
  booktitle = {Proceedings of NAACL 2024},
  year      = {2024}
}

@misc{atreja2026pioneeragent,
  title  = {Pioneer Agent: Continual Improvem
  author = {Atreja, Dhruv and White, Julia an
  year   = {2026},
  url    = {https://arxiv.org/abs/2604.09791}
}
```


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